

DPP-5

- Q.1 If $\vec{e}_1$ & $\vec{e}_2$ are two unit vectors and θ is the angle between them, then $\sin(\theta/2)$ is :
- (A) $\frac{1}{2} |\vec{e}_1 + \vec{e}_2|$ (B) $\frac{1}{2} |\vec{e}_1 - \vec{e}_2|$ (C) $\frac{\vec{e}_1 \cdot \vec{e}_2}{2}$ (D) $\frac{|\vec{e}_1 \times \vec{e}_2|}{2|\vec{e}_1||\vec{e}_2|}$
- Q.2 If $\vec{p}$ & $\vec{s}$ are not perpendicular to each other and $\vec{r} \times \vec{p} = \vec{q} \times \vec{p}$ & $\vec{r} \cdot \vec{s} = 0$, then $\vec{r} =$
- (A) $\vec{p} \cdot \vec{s}$ (B) $\vec{q} + \left(\frac{\vec{q} \cdot \vec{p}}{\vec{p} \cdot \vec{s}} \right) \vec{p}$ (C) $\vec{q} - \left(\frac{\vec{q} \cdot \vec{s}}{\vec{p} \cdot \vec{s}} \right) \vec{p}$ (D) $\vec{q} + \mu \vec{p}$ for all scalars μ
- Q.3 If $\vec{u} = \vec{a} - \vec{b}$; $\vec{v} = \vec{a} + \vec{b}$ and $|\vec{a}| = |\vec{b}| = 2$ then $|\vec{u} \times \vec{v}|$ is equal to
- (A) $\sqrt{2(16 - (\vec{a} \cdot \vec{b})^2)}$ (B) $2\sqrt{16 - (\vec{a} \cdot \vec{b})^2}$ (C) $2\sqrt{4 - (\vec{a} \cdot \vec{b})^2}$ (D) $\sqrt{2(4 - (\vec{a} \cdot \vec{b})^2)}$
- Q.4 If $\vec{u}$ and $\vec{v}$ are two vectors such that $|\vec{u}| = 3$; $|\vec{v}| = 2$ and $|\vec{u} \times \vec{v}| = 6$ then the correct statement is
- (A) $\vec{u} \wedge \vec{v} \in (0, 90^\circ)$ (B) $\vec{u} \wedge \vec{v} \in (90^\circ, 180^\circ)$ (C) $\vec{u} \wedge \vec{v} = 90^\circ$ (D) $(\vec{u} \times \vec{v}) \times \vec{u} = 6\vec{v}$
- Q.5 If $\vec{A} = (1, 1, 1)$, $\vec{C} = (0, 1, -1)$ are given vectors, then a vector $\vec{B}$ satisfying the equation $\vec{A} \times \vec{B} = \vec{C}$ and $\vec{A} \cdot \vec{B} = 3$ is :
- (A) $(5, 2, 2)$ (B) $\left(\frac{5}{3}, \frac{2}{3}, \frac{2}{3}\right)$ (C) $\left(\frac{2}{3}, \frac{5}{3}, \frac{2}{3}\right)$ (D) $\left(\frac{2}{3}, \frac{2}{3}, \frac{5}{3}\right)$
- Q.6 Given a parallelogram OACB. The lengths of the vectors $\vec{OA}$, $\vec{OB}$ & $\vec{AB}$ are a , b & c respectively. The scalar product of the vectors $\vec{OC}$ & $\vec{OB}$ is :
- (A) $\frac{a^2 - 3b^2 + c^2}{2}$ (B) $\frac{3a^2 + b^2 - c^2}{2}$ (C) $\frac{3a^2 - b^2 + c^2}{2}$ (D) $\frac{a^2 + 3b^2 - c^2}{2}$
- Q.7 Vectors $\vec{a}$ & $\vec{b}$ make an angle $\theta = \frac{2\pi}{3}$. If $|\vec{a}| = 1$, $|\vec{b}| = 2$ then $\left\{ (\vec{a} + 3\vec{b}) \times (3\vec{a} - \vec{b}) \right\}^2 =$
- (A) 225 (B) 250 (C) 275 (D) 300
- Q.8 In a quadrilateral ABCD, $\vec{AC}$ is the bisector of the $(\vec{AB} \wedge \vec{AD})$ which is $\frac{2\pi}{3}$,
 $15|\vec{AC}| = 3|\vec{AB}| = 5|\vec{AD}|$ then $\cos(\vec{BA} \wedge \vec{CD})$ is :
- (A) $-\frac{\sqrt{14}}{7\sqrt{2}}$ (B) $-\frac{\sqrt{21}}{7\sqrt{3}}$ (C) $\frac{2}{\sqrt{7}}$ (D) $\frac{2\sqrt{7}}{14}$

- Q.9 If the two adjacent sides of two rectangles are represented by the vectors $\vec{p} = 5\vec{a} - 3\vec{b}$; $\vec{q} = -\vec{a} - 2\vec{b}$ and $\vec{r} = -4\vec{a} - \vec{b}$; $\vec{s} = -\vec{a} + \vec{b}$ respectively, then the angle between the vectors $\vec{x} = \frac{1}{3}(\vec{p} + \vec{r} + \vec{s})$ and $\vec{y} = \frac{1}{5}(\vec{r} + \vec{s})$
- (A) is $-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (B) is $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$
- (C) is $\pi - \cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ (D) cannot be evaluated
- Q.10 If the vector product of a constant vector $\vec{OA}$ with a variable vector $\vec{OB}$ in a fixed plane OAB be a constant vector, then locus of B is :
- (A) a straight line perpendicular to $\vec{OA}$ (B) a circle with centre O radius equal to $|\vec{OA}|$
- (C) a straight line parallel to $\vec{OA}$ (D) none of these
- Q.11 If the distance from the point P(1, 1, 1) to the line passing through the points Q(0, 6, 8) and R(-1, 4, 7) is expressed in the form $\sqrt{p/q}$ where p and q are coprime, then the value of $\frac{(p+q)(p+q-1)}{2}$ equals
- (A) 4950 (B) 5050 (C) 5150 (D) none